

Week 2

Optimization Principles for Federated Learning

From empirical risk to federated objectives

Theory and Applications of Federated Learning

Graduate course · English instruction

Faranak Solat

Department of AI & Data Science

Sejong University



Learning Objectives

Optimization language

Express supervised learning as empirical risk minimization and identify model parameters, losses, and gradients.

Gradient-based learning

Distinguish batch gradient descent, stochastic gradient descent, and mini-batch SGD.

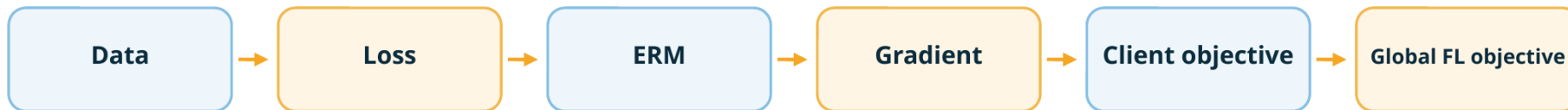
Federated objective

Connect client-level objectives to the weighted global FL objective.

Convergence intuition

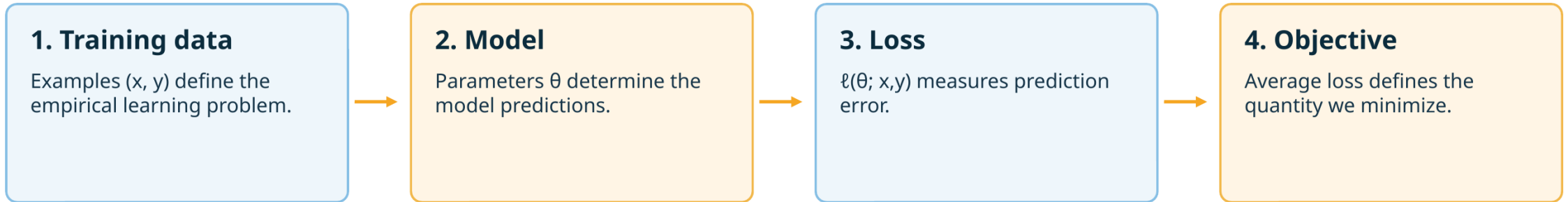
Reason about step size, stochasticity, local steps, and why FL optimization is harder than centralized training.

Today's path

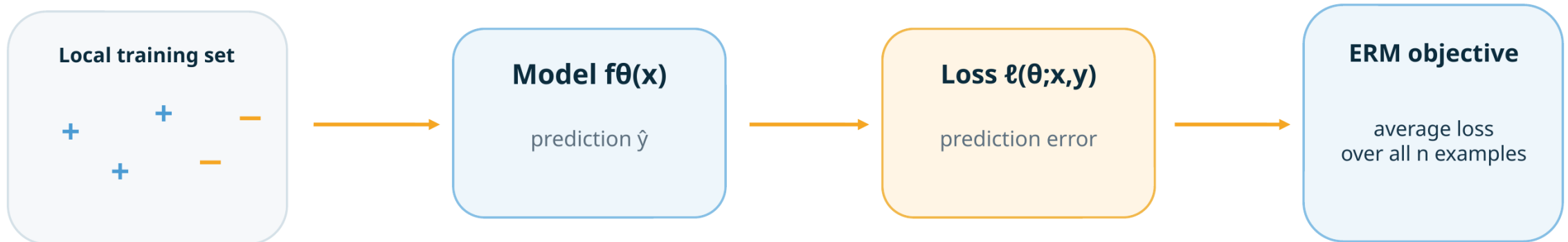


Goal: move from ordinary ML optimization to the objective that FedAvg will optimize next week.

From Data to an Optimization Problem



Example: binary classifier



Training is an optimization problem once the model, loss, and objective are specified.

Empirical Risk Minimization

$$\text{minimize over } \theta: F(\theta) = (1/n) \sum_i \ell(\theta; \mathbf{x}_i, y_i)$$

ERM turns model training into an optimization problem: choose parameters that minimize average training loss.

θ — model parameters

The weights and biases that training adjusts.

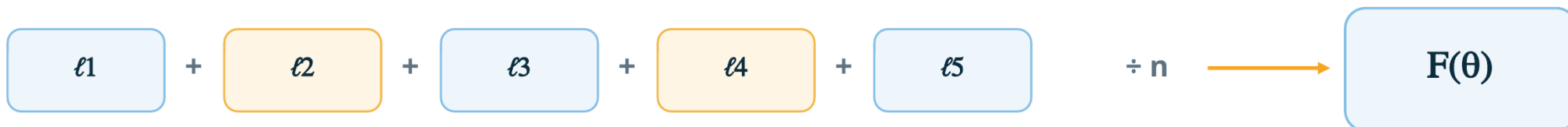
ℓ — per-example loss

Measures how wrong the model is on one example.

$F(\theta)$ — empirical objective

Average loss across the complete training set.

ERM = average of individual losses



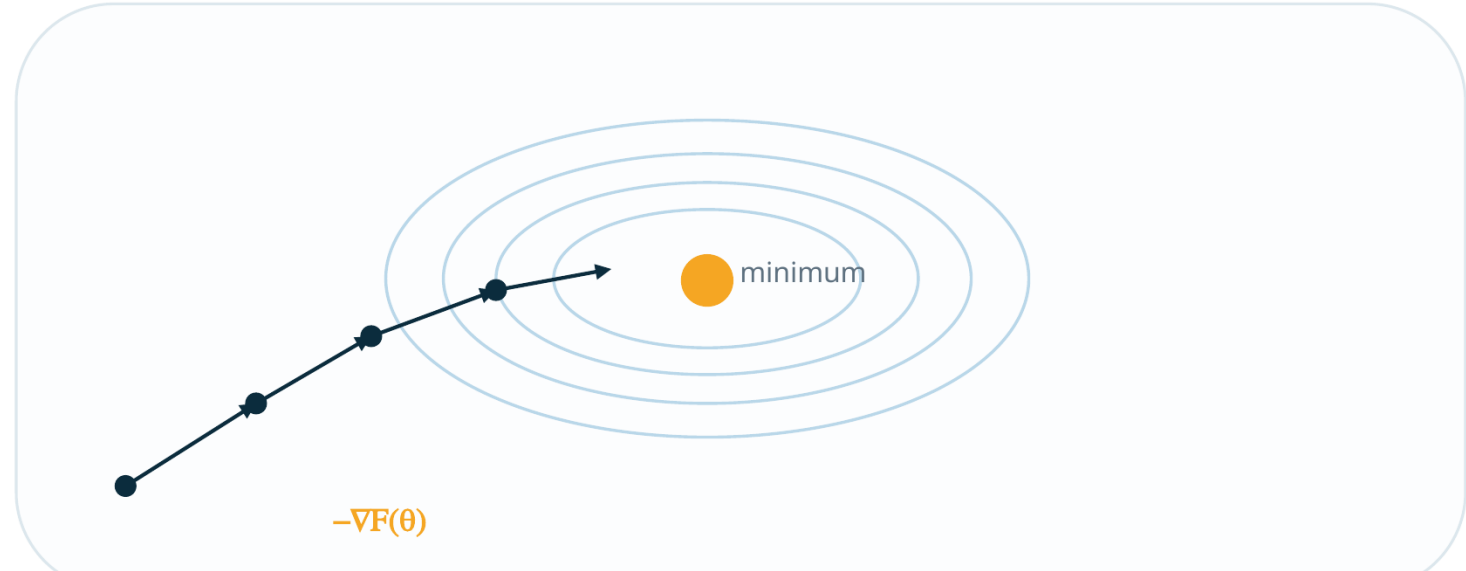
In FL, the same ERM idea will be expressed as a weighted sum of client-level objectives.

Loss Functions and Optimization Landscapes

Optimization is a search over parameter space

Gradient intuition

The gradient points in the direction of steepest increase. Moving in the negative-gradient direction reduces the objective locally.



Classification loss

Cross-entropy / logistic loss turns prediction error into a differentiable objective.

Regression loss

Squared error measures the distance between predictions and continuous targets.

Landscape matters

Curvature and non-convexity determine how easy it is to optimize and analyze convergence.

A gradient is local information: it tells us how to move now, not where the global optimum is.

Gradient Descent

$$\theta_{t+1} = \theta_t - \eta \nabla F(\theta_t)$$

Gradient descent repeatedly moves the parameters in a direction that locally decreases the objective.

η — learning rate

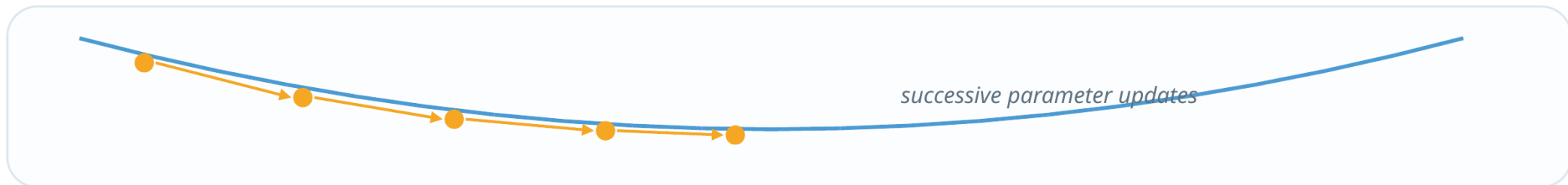
Controls the step size. Too large may diverge; too small may learn very slowly.

$\nabla F(\theta)$ — gradient

Encodes the local direction and magnitude of change.

Iteration

Repeat update steps until a stopping condition is met.

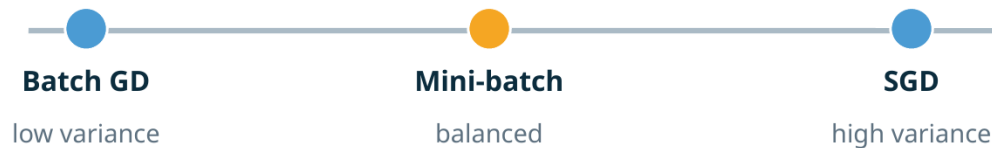


Central question for FL: how accurately can distributed clients estimate and apply the global descent direction?

Batch GD, SGD, and Mini-Batch SGD

Method	Gradient estimate	Update frequency	Strength	Limitation
Batch GD	All training examples	Once per full dataset	Stable direction	Expensive for large datasets
SGD	One example	Very frequent	Cheap, noisy exploration	High variance
Mini-batch SGD	Small batch	Frequent and parallelizable	Efficient + scalable	Batch size is a tuning choice

How noisy is the gradient estimate?



Why this matters for federated learning

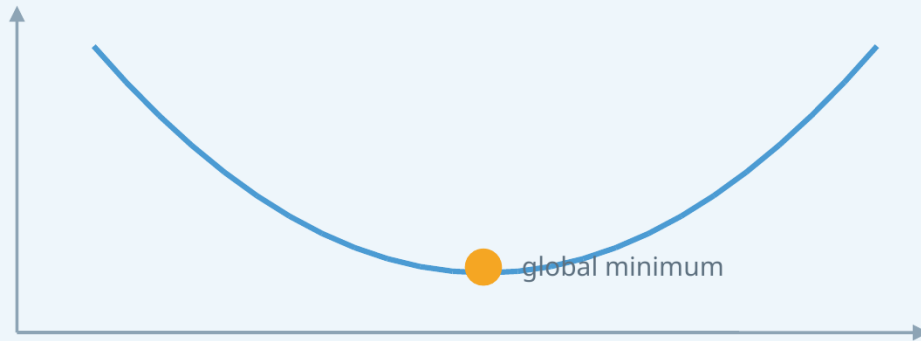
- Clients usually run mini-batch SGD locally.
- Different clients see different mini-batches and different data distributions.
- Local stochastic noise combines with cross-client heterogeneity.

FedAvg is essentially local mini-batch SGD plus periodic model averaging across clients.

Convex vs. Non-Convex Optimization

Convex objectives

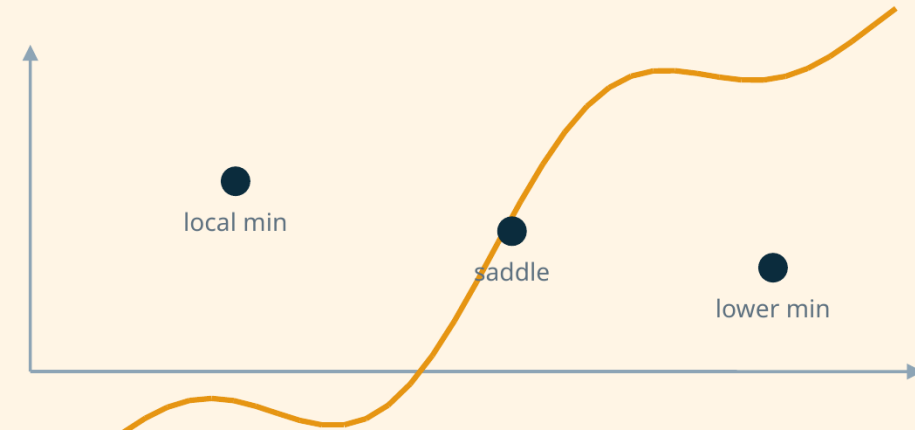
- Any local minimum is also a global minimum.
- The objective has no deceptive local basins.
- Convex analysis gives clean convergence guarantees.



Useful theoretical model

Non-convex objectives

- Deep neural networks create non-convex loss surfaces.
- There may be local minima, flat regions, and saddle points.
- We typically seek stationary points rather than guaranteed global optima.



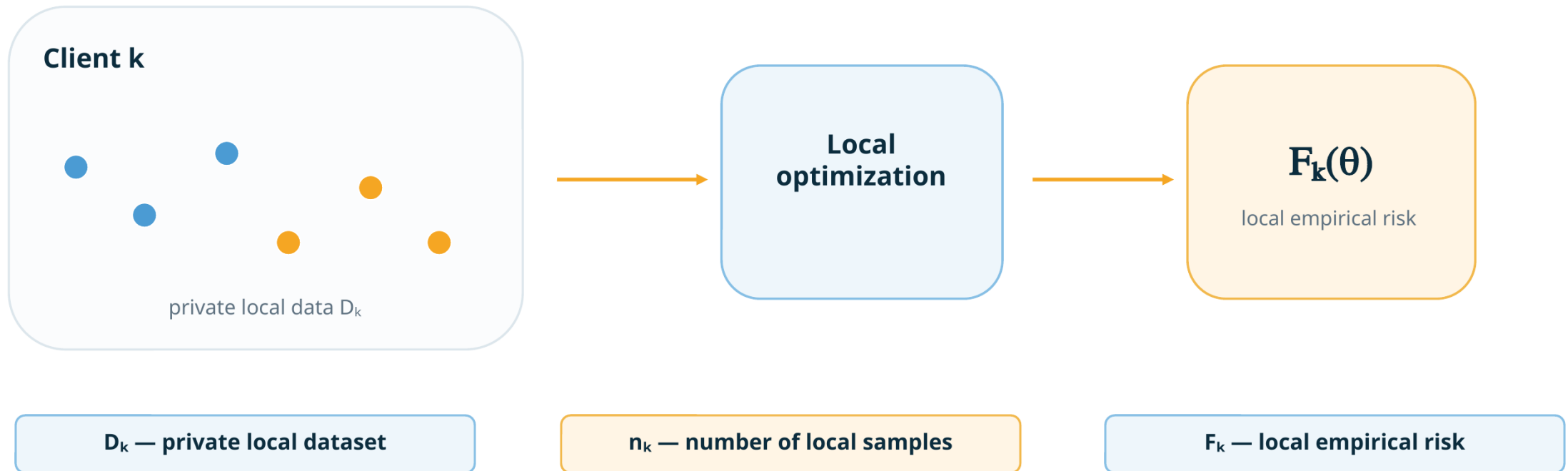
Most FL with neural networks is non-convex

Convex theory helps us reason rigorously; practical deep FL usually operates in the non-convex regime.

Client-Level Objective

$$F_k(\theta) = (1/n_k) \sum_{i \in D_k} \ell(\theta; \mathbf{x}_i, \mathbf{y}_i)$$

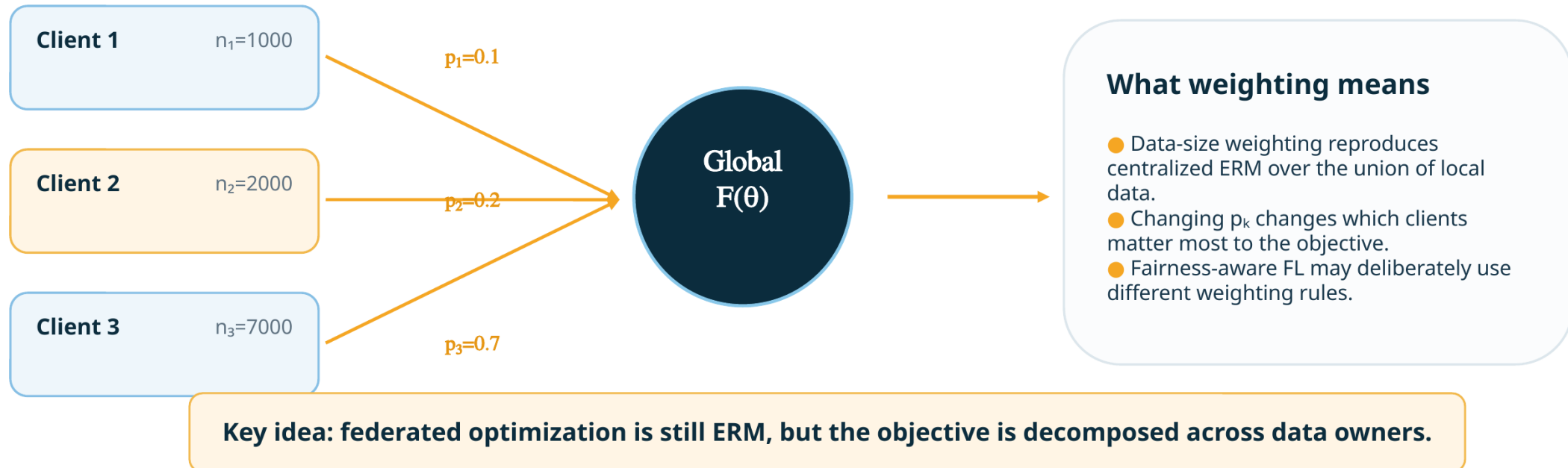
Client k can compute its objective using only its local dataset D_k .



Global Federated Objective

$$F(\theta) = \sum_k p_k F_k(\theta), \quad \text{typically } p_k = n_k / \sum_j n_j$$

The global objective combines client objectives without centralizing raw data.



Why Federated Optimization Is Different

Non-IID data

Client objectives may point in different gradient directions.

Unbalanced data

Clients can hold very different numbers of samples.

Partial participation

Only a subset of clients may be available in a round.

Communication limits

Model exchange is much slower than local arithmetic.

System heterogeneity

Compute speed, memory, battery, and connectivity differ.

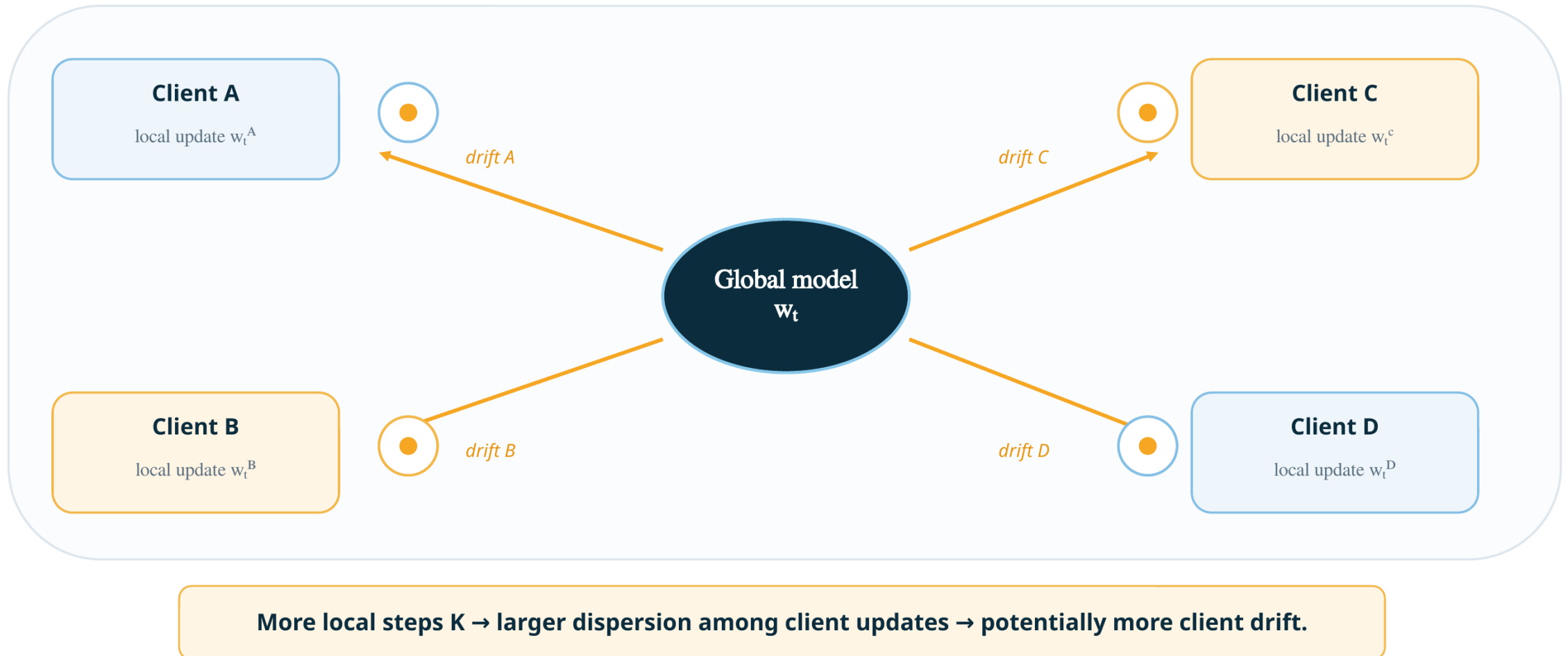
Privacy mechanisms

Secure aggregation or DP can constrain algorithm design.

In FL, the optimization algorithm must coexist with communication, privacy, and systems constraints.

Client Drift: The Central FL Optimization Tension

Local optimization can pull clients toward different local minima



Basic Convergence Intuition

Objective decrease

Training should reduce the target objective or gradient norm over time.

Learning rate η

Too large can diverge; too small can learn extremely slowly.

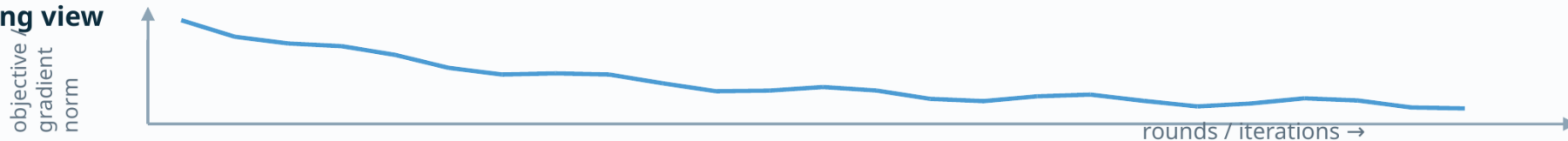
Stochastic noise

Mini-batches add variance but make training scalable.

Local steps K

More local computation saves communication but can increase model divergence.

Typical training view



Convergence in FL is a balance among step size, stochasticity, heterogeneity, local work, and communication.

Computation–Communication Trade-off

One knob: the number of local steps K between aggregations



More frequent aggregation

Local models stay closer to one another and to the global trajectory.

- Lower drift risk
- More synchronization
- Higher communication cost

More local computation

Fewer communication rounds can reduce wall-clock cost, but local models move farther toward their own objectives.

- Lower communication cost
- More local work
- Higher drift risk

What changes as K increases?

Communication rounds ↓

Client drift risk ↑

Local compute / round ↑

Synchronization overhead ↓

In-Class Activity: Build the Federated Objective

Scenario

Three hospitals train the same classifier. They have 1,000, 2,000, and 7,000 local records.

Hospital A		1,000
Hospital B		2,000
Hospital C		7,000

Task 1

Write each hospital local empirical risk $F_k(\theta)$.

Task 2

Choose p_k so that the federated objective matches centralized ERM.

Task 3

Discuss what changes if one hospital has a very different patient distribution.

Expected weights

$$p_1 = 0.1 \quad p_2 = 0.2 \quad p_3 = 0.7$$

Discussion: data-size weighting matches centralized ERM, but it does not solve non-IID data or fairness.

Key Takeaways

What to remember before leaving today

- 1 Machine learning training can be framed as empirical risk minimization.
- 2 Gradient descent, SGD, and mini-batch SGD differ mainly in how they estimate the gradient.
- 3 Federated learning optimizes a weighted combination of client-level objectives.
- 4 Non-IID data, partial participation, system constraints, and communication fundamentally change the optimization problem.
- 5 Next week: turn these principles into the FedAvg training algorithm.

Mental model: FL optimization = local stochastic optimization + periodic aggregation + heterogeneity.

Reading & Preparation

Suggested reading before the next session

Main textbook

Ludwig & Baracaldo (2022)
Chapter 6: Communication-Efficient Distributed Optimization Algorithms.

Supplementary reference

Kairouz et al. (2021)
Section 3.2: Optimization Algorithms for Federated Learning.

Prepare for Week 3 — FedAvg

- Review local SGD and weighted averaging.
- Be able to explain one communication round.
- Think about what changes when only a subset of clients participates.

Next week we will connect these pieces into the full FedAvg loop.

Preparation focus: local SGD, weighted averaging, client sampling, and communication rounds.

Q & A

Thank You.

sofar@sejong.ac.kr